有限惯性代数栈的模空间装配

Counting scope. The primary problem is Tag 0DUT, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

Copyright (C) 2005–2025 Johan de Jong. Stacks Project authors. Source: Tag 0DUT.

Difficulty H3: 在给定 étale 局部有限惯性覆盖及 well-nigh-affine 局部模空间后，仍须构造整体模空间与模射。证明自行组成 M_ij 的 étale 等价关系、验证几何点上的关系并粘合 f，再核实基本图表为笛卡尔及任意平坦基变换后的范畴泛性质。局部存在性是被引输入，整体存在性与泛性质装配仍有独立实质内容，因此列为 H3 候选。.

History: excerpt from revision local cached snapshot; exact chapter hashes; revision unknown, stacks-more-morphisms.tex, lines 3636–3885. Only reference presentation was adapted. Licensed under GNU FDL 1.2 or later; no invariant sections or cover texts. See accompanying COPYING.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Assume $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is finite. Then there exist a set I and for $i \in I$ a morphism of algebraic stacks*

$$g_i : \mathcal{X}_i \longrightarrow \mathcal{X}$$

with the following properties

(1) $|\mathcal{X}| = \bigcup |g_i|(|\mathcal{X}_i|)$,
(2) *$\mathcal{X}_i$ is well-nigh affine,*
(3) *$\mathcal{I}_{\mathcal{X}_i} \to \mathcal{X}_i \times_\mathcal{X} \mathcal{I}_\mathcal{X}$ is an isomorphism, and*
(4) *$g_i : \mathcal{X}_i \to \mathcal{X}$ is representable by algebraic spaces, separated, and étale,*

证明. For any $x \in |\mathcal{X}|$ we can choose $g : \mathcal{U} \to \mathcal{X}, \mathcal{U} = [U/R]$, and u as in Morphisms of Stacks, Lemma lemma etale local quasi DM at x (Tag 0DU4). Then by Morphisms of Stacks, Lemma lemma stabilizer preserving unramified (Tag 0DUA) we see that there exists an open substack $\mathcal{U}' \subset \mathcal{U}$ containing u such that $\mathcal{I}_{\mathcal{U}'} \to \mathcal{U}' \times_\mathcal{X} \mathcal{I}_\mathcal{X}$ is an isomorphism. Let $U' \subset U$ be the R-invariant open corresponding to the open substack $\mathcal{U}'$. Let $u' \in U'$ be a point of U' mapping to u. Observe that $t(s^{-1}(\{u'\}))$ is finite as $s : R \to U$ is finite. By Properties, Lemma lemma ample finite set in affine (Tag 01ZY) and Groupoids, Lemma lemma find invariant affine (Tag 03JE) we can find an R-invariant affine open $U'' \subset U'$ containing u'. Let R'' be the restriction of R to U''. Then $\mathcal{U}'' = [U''/R'']$ is an open substack of $\mathcal{U}'$ containing u, is well-nigh affine, $\mathcal{I}_{\mathcal{U}''} \to \mathcal{U}'' \times_\mathcal{X} \mathcal{I}_\mathcal{X}$ is an isomorphism, and $\mathcal{U}'' \to \mathcal{X}$ and is representable by algebraic spaces and étale. Finally, $\mathcal{U}'' \to \mathcal{X}$ is separated as $\mathcal{U}''$ is separated (Lemma lemma well nigh affine (Tag 0DUM)) the diagonal of $\mathcal{X}$ is separated (Morphisms of Stacks, Lemma lemma diagonal diagonal (Tag 0CL0)) and separatedness follows from Morphisms of Stacks, Lemma lemma compose after separated (Tag 050M). Since the point $x \in |\mathcal{X}|$ is arbitrary the proof is complete. □

Lemma 0.2. *Let the algebraic stack $\mathcal{X}$ be well-nigh affine. There exists a uniform categorical moduli space*

$$f : \mathcal{X} \longrightarrow M$$

in the category of affine schemes. Moreover f is separated, quasi-compact, and a universal homeomorphism.

证明. Write $\mathcal{X} = [U/R]$ with (U, R, s, t, c) as in Lemma lemma well nigh affine (Tag 0DUM). Let C be the ring of R-invariant functions on U, see Groupoids, Section section finite flat (Tag 03BE). We set $M = \mathrm{Spec}(C)$. The R-invariant morphism $U \to M$ corresponds to a morphism $f : \mathcal{X} \to M$ by Lemma lemma quotient compare (Tag 0DUH). The characterization of morphisms into affine schemes given in Schemes, Lemma lemma morphism into affine (Tag 01I1) immediately guarantees that $\phi : U \to M$ is a categorical quotient in the category of affine schemes. Hence f is a categorical moduli space in the category of affine schemes (Lemma lemma categorical quotient compare (Tag 0DUI)).

Since $\mathcal{X}$ is separated by Lemma lemma well nigh affine (Tag 0DUM) we find that f is separated by Morphisms of Stacks, Lemma lemma compose after separated (Tag 050M).

Since $U \to \mathcal{X}$ is surjective and since $U \to M$ is quasi-compact, we see that f is quasi-compact by Morphisms of Stacks, Lemma lemma surjection from quasi compact (Tag 050X).

By Groupoids, Lemma lemma integral over invariants (Tag 03BJ) the composition

$$U \to \mathcal{X} \to M$$

is an integral morphism of affine schemes. In particular, it is universally closed (Morphisms, Lemma lemma integral universally closed (Tag 01WM)). Since $U \to \mathcal{X}$ is surjective, it follows that $\mathcal{X} \to M$ is universally closed (Morphisms of Stacks, Lemma lemma image proper is proper (Tag 0CQK)). To conclude that $\mathcal{X} \to M$ is a universal homeomorphism, it is enough to show that it is universally bijective, i.e., surjective and universally injective.

We have $|\mathcal{X}| = |U|/|R|$ by Morphisms of Stacks, Lemma lemma points presentation (Tag 0DTZ). Thus $|f|$ is surjective and even bijective by Groupoids, Lemma lemma points (Tag 03BL).

Let $C \to C'$ be a ring map. Let (U', R', s', t', c') be the base change of (U, R, s, t, c) by $M' = \mathrm{Spec}(C') \to M$. Setting $\mathcal{X}' = [U'/R']$, we observe that $M' \times_M \mathcal{X} = \mathcal{X}'$ by Quotients of Groupoids, Lemma lemma base change quotient stack (Tag 0DTG). Let C^1 be the ring of R'-invariant functions on U'. Set $M^1 = \mathrm{Spec}(C^1)$ and consider the diagram

$$\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{X} \\ \downarrow{\scriptstyle f'} & & \\ M^1 & & \downarrow{\scriptstyle f} \\ \downarrow & & \\ M' & \longrightarrow & M \end{array}$$

By Groupoids, Lemma lemma invariants base change (Tag 03BK) and Algebra, Lemma lemma universally bijective (Tag 09EF) the morphism $M^1 \to M'$ is a homeomorphism. On the other hand, the previous paragraph applied to (U', R', s', t', c') shows that $|f'|$ is bijective. We conclude that f induces a bijection on points after any base change by an affine scheme. Thus f is universally injective by Morphisms of Stacks, Lemma lemma universally injective local (Tag 0DTN).

Finally, we still have to show that f is a uniform moduli space in the category of affine schemes. This follows from the discussion above and the fact that if the ring map $C \to C'$ is flat, then $C' \to C^1$ is an isomorphism by Groupoids, Lemma lemma invariants base change (Tag 03BK). □

Lemma 0.3. *Let $h : \mathcal{X}' \to \mathcal{X}$ be a morphism of algebraic stacks. Assume $\mathcal{X}$ is well-nigh affine, h is étale, h is separated, and h induces isomorphisms on automorphism groups (Morphisms of Stacks, Remark remark identify automorphism groups (Tag 0DTW)). Then there exists a cartesian diagram*

$$\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ M' & \longrightarrow & M \end{array}$$

where $M' \to M$ is a separated étale morphism of schemes and $\mathcal{X} \to M$ is the moduli space constructed in Lemma 0.2.

证明. Choose an affine scheme U and a surjective, flat, finite, and locally finitely presented morphism $U \to \mathcal{X}$. Since h is representable by algebraic spaces (Morphisms of Stacks, Lemmas lemma aut iso unramified (Tag 0DU9) and lemma stabilizer preserving (Tag 0DU7)) we see that $U' = \mathcal{X}' \times_{\mathcal{X}} U$ is an algebraic space. Since $U' \to U$ is separated and étale, we see that U' is a scheme and that every finite set of points of U' is contained in an affine open, see Morphisms of Spaces, Lemma lemma locally quasi finite separated representable (Tag 0418) and More on Morphisms, Lemma lemma separated locally quasi finite over affine (Tag 07S0). Setting $R' = U' \times_{\mathcal{X}'} U'$ we see that $s', t' : R' \to U'$ are finite locally free. By Groupoids, Lemma lemma find invariant affine (Tag 03JE) there exists an open covering $U' = \bigcup U'_i$ by R'-invariant affine open subschemes $U'_i \subset U'$. Let $\mathcal{X}'_i \subset \mathcal{X}'$ be the corresponding open substacks. These are well-nigh affine as $U'_i \to \mathcal{X}'_i$ is surjective, flat, finite and of finite presentation. By Lemma lemma well nigh affine moduli space etale (Tag 0DUQ) we obtain cartesian diagrams

$$\begin{array}{ccc} \mathcal{X}'_i & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ M'_i & \longrightarrow & M \end{array}$$

with $M'_i \to M$ an étale morphism of affine schemes and vertical arrows as in Lemma 0.2. Observe that $\mathcal{X}'_{ij} = \mathcal{X}'_i \cap \mathcal{X}'_j$ is an open subspace of $\mathcal{X}'_i$ and $\mathcal{X}'_j$. Hence we get corresponding open subschemes $V_{ij} \subset M'_i$ and $V_{ji} \subset M'_j$. By the result of Lemma lemma moduli space finite affine (Tag 0DUR) we see that both $\mathcal{X}'_{ij} \to V_{ij}$ and $\mathcal{X}'_{ji} \to V_{ji}$ are categorical moduli spaces! Thus we get a unique isomorphism $\varphi_{ij} : V_{ij} \to V_{ji}$ such that

$$\begin{array}{ccccccc} \mathcal{X}'_i & \longleftarrow & \mathcal{X}'_i \cap \mathcal{X}'_j & & \longrightarrow & & \mathcal{X}'_j \\ \downarrow & & \swarrow & & \searrow & & \downarrow \\ M'_i & \longleftarrow & V_{ij} & \xrightarrow{\varphi_{ij}} & V_{ji} & \longrightarrow & M'_j \end{array}$$

is commutative. These isomorphisms satisfy the cocyclce condition of Schemes, Section section glueing schemes (Tag 01JA) by a computation (and another application of the previous lemma) which we omit. Thus we can glue the affine schemes in to scheme M', see Schemes, Lemma lemma glue (Tag 01JB). Let us identify the M'_i with their image in M'. We claim there is a morphism $\mathcal{X}' \to M'$ fitting into cartesian diagrams

$$\begin{array}{ccc} \mathcal{X}'_i & \longrightarrow & \mathcal{X}' \\ \downarrow & & \downarrow \\ M'_i & \longrightarrow & M' \end{array}$$

This is clear from the description of the morphisms into the glued scheme M' in Schemes, Lemma lemma glue (Tag 01JB) and the fact that to give a morphism $\mathcal{X}' \to M'$ is the same thing as given a morphism $T \to M'$ for any morphism $T \to \mathcal{X}'$. Similarly, there is a morphism $M' \to M$ restricting to the given morphisms $M'_i \to M$ on M'_i. The morphism $M' \to M$ is étale (being étale on the members of an étale covering) and the fibre product property holds as it can be checked on members of the (affine) open covering $M' = \bigcup M'_i$. Finally, $M' \to M$ is separated because the composition $U' \to \mathcal{X}' \to M'$ is surjective and universally closed and we can apply Morphisms, Lemma lemma image universally closed separated (Tag 09MQ). □

Primary problem: Tag 0DUT

Theorem 0.4 (Keel-Mori)**.** *Let $\mathcal{X}$ be an algebraic stack. Assume $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is finite. Then there exists a uniform categorical moduli space*

$$f : \mathcal{X} \longrightarrow M$$

and f is separated, quasi-compact, and a universal homeomorphism.

证明. We choose a set I[1] and for $i \in I$ a morphism of algebraic stacks $g_i : \mathcal{X}_i \to \mathcal{X}$ as in Lemma 0.1; we will use all of the properties listed in this lemma without further mention. Let

$$f_i : \mathcal{X}_i \to M_i$$

be as in Lemma 0.2. Consider the stacks

$$\mathcal{X}_{ij} = \mathcal{X}_i \times_{g_i,\mathcal{X},g_j} \mathcal{X}_j$$

for $i, j \in I$. The projections $\mathcal{X}_{ij} \to \mathcal{X}_i$ and $\mathcal{X}_{ij} \to \mathcal{X}_j$ are separated by Morphisms of Stacks, Lemma lemma base change separated (Tag 050F), étale by Morphisms of Stacks, Lemma lemma base change etale (Tag 0CIN), and induce isomorphisms on automorphism groups (as in Morphisms of Stacks, Remark remark identify automorphism groups (Tag 0DTW)) by Morphisms of Stacks, Lemma lemma base change stabilizer preserving (Tag 0DUB). Thus we may apply Lemma 0.3 to find a commutative diagram

$$\begin{array}{ccccc} \mathcal{X}_i & \longleftarrow & \mathcal{X}_{ij} & \longrightarrow & \mathcal{X}_j \\ {\scriptstyle f_i}\downarrow & & {\scriptstyle f_{ij}}\downarrow & & {\scriptstyle f_j}\downarrow \\ M_i & \longleftarrow & M_{ij} & \longrightarrow & M_j \end{array}$$

with cartesian squares where $M_{ij} \to M_i$ and $M_{ij} \to M_j$ are separated étale morphisms of schemes; here we also use that f_i is a uniform categorical quotient by Lemma lemma moduli space finite affine (Tag 0DUR). Claim:

$$\coprod M_{ij} \longrightarrow \coprod M_i \times \coprod M_i$$

is an étale equivalence relation.

Proof of the claim. Set $R = \coprod M_{ij}$ and $U = \coprod M_i$. We have already seen that $t : R \to U$ and $s : R \to U$ are étale. Let us construct a morphism $c : R \times_{s,U,t} R \to R$ compatible with $\mathrm{pr}_{13} : U \times U \times U \to U \times U$. Namely, for $i, j, k \in I$ we consider

$$\mathcal{X}_{ijk} = \mathcal{X}_i \times_{g_i,\mathcal{X},g_j} \mathcal{X}_j \times_{g_j,\mathcal{X},g_k} \mathcal{X}_k = \mathcal{X}_{ij} \times_{\mathcal{X}_j} \mathcal{X}_{jk}$$

[1]The reader who is still keeping track of set theoretic issues should make sure I is not too large.

Arguing exactly as in the previous paragraph, we find that $M_{ijk} = M_{ij} \times_{M_j} M_{jk}$ is a categorical moduli space for $\mathcal{X}_{ijk}$. In particular, there is a canonical morphism $M_{ijk} = M_{ij} \times_{M_j} M_{jk} \to M_{ik}$ coming from the projection $\mathcal{X}_{ijk} \to \mathcal{X}_{ik}$. Putting these morphisms together we obtain the morphism c. In a similar fashion we construct a morphism $e : U \to R$ compatible with $\Delta : U \to U \times U$ and $i : R \to R$ compatible with the flip $U \times U \to U \times U$. Let k be an algebraically closed field. Then

$$\mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X}_i) \to \mathrm{Mor}(\mathrm{Spec}(k), M_i) = M_i(k)$$

is bijective on isomorphism classes and the same remains true after any base change by a morphism $M' \to M$. This follows from our choice of f_i and Morphisms of Stacks, Lemmas lemma universally injective (Tag 0CPN) and lemma universally injective point (Tag 0DTM). By construction of 2-fibred products the diagram

$$\begin{array}{ccc} \mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X}_{ij}) & \longrightarrow & \mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X}_j) \\ \downarrow & & \downarrow \\ \mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X}_i) & \longrightarrow & \mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X}) \end{array}$$

is a fibre product of categories. By our choice of g_i the functors in this diagram induce bijections on automorphism groups. It follows that this diagram induces a fibre product diagram on sets of isomorphism classes! Thus we see that

$$R(k) = U(k) \times_{|\mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X})|} U(k)$$

where $|\mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X})|$ denotes the set of isomorphism classes. In particular, for any algebraically closed field k the map on k-valued point is an equivalence relation. We conclude the claim holds by Groupoids, Lemma lemma etale equivalence relation (Tag 0DT7).

Let $M = U/R$ be the algebraic space which is the quotient of the above étale equivalence relation, see Spaces, Theorem theorem presentation (Tag 02WW). There is a canonical morphism $f : \mathcal{X} \to M$ fitting into commutative diagrams

$$\begin{array}{ccc} \mathcal{X}_i & \xrightarrow[g_i]{} & \mathcal{X} \\ {\scriptstyle f_i}\downarrow & & \downarrow{\scriptstyle f} \\ M_i & \longrightarrow & M \end{array} \tag{0.4.1}$$

Namely, such a morphism f is given by a functor

$$f : \mathrm{Mor}(T, \mathcal{X}) \longrightarrow \mathrm{Mor}(T, M)$$

for any scheme T compatible with base change. Let $a : T \to \mathcal{X}$ be an object of the left hand side. We obtain an étale covering $\{T_i \to T\}$ with $T_i = \mathcal{X}_i \times_{\mathcal{X}} T$ and morphisms $a_i : T_i \to \mathcal{X}_i$. Then we get $b_i = f_i \circ a_i : T_i \to M_i$. Since $T_i \times_T T_j = \mathcal{X}_{ij} \times_{\mathcal{X}} T$ we moreover get a morphism $a_{ij} : T_i \times_T T_j \to \mathcal{X}_{ij}$. Setting $b_{ij} = f_{ij} \circ a_{ij}$ we find that $b_i \times b_j$ factors through the monomorphism $M_{ij} \to M_i \times M_j$. Hence the morphisms

$$T_i \xrightarrow{b_i} M_i \to M$$

agree on $T_i \times_T T_j$. As M is a sheaf for the étale topology, we see that these morphisms glue to a unique morphism $b = f(a) : T \to M$. We omit the verification that this construction is compatible with base change and we omit the verification that the diagrams (0.4.1) commute.

Claim: the diagrams (0.4.1) are cartesian. To see this we study the induced morphism

$$h_i : \mathcal{X}_i \longrightarrow M_i \times_M \mathcal{X}$$

This is a morphism of stacks étale over $\mathcal{X}$ and hence h_i is étale (Morphisms of Stacks, Lemma lemma etale permanence (Tag 0CIR)). Since g_i is separated, we see h_i is separated (use Morphisms of Stacks, Lemma lemma compose after separated (Tag 050M) and the fact seen above that the diagonal of $\mathcal{X}$ is separated). The morphism h_i induces isomorphisms on automorphism groups (Morphisms of Stacks, Remark remark identify automorphism groups (Tag 0DTW)) as this is true for g_i. For an algebraically closed field k the diagram

$$\begin{array}{ccc} \mathrm{Mor}(\mathrm{Spec}(k), M_i \times_M \mathcal{X}) & \longrightarrow & \mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X}) \\ \downarrow & & \downarrow \\ M_i(k) & \longrightarrow & M(k) \end{array}$$

is a catesian diagram of categories and the top arrow induces bijections on automorphism groups. On the other hand, we have

$$M(k) = U(k)/R(k) = U(k)/U(k) \times_{|\mathrm{Mor}(\mathrm{Spec}(k),\mathcal{X})|} U(k) = |\mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X})|$$

by what we said above. Thus the right vertical arrow in the cartesian diagram above is a bijection on isomorphism classes. We conclude that $|\mathrm{Mor}(\mathrm{Spec}(k), M_i \times_M \mathcal{X})| \to M_i(k)$ is bijective. Review: h_i is a separated, étale, induces isomorphisms on automorphism groups (as in Morphisms of Stacks, Remark remark identify automorphism groups (Tag 0DTW)), and induces an equivalence on fibre categories over algebraically closed fields. Hence it is an isomorphism by Morphisms of Stacks, Lemma lemma etale iso (Tag 0DUD).

From the claim we get in particular the following: we have a surjective étale morphism $U \to M$ such that the base change of f is separated, quasi-compact, and a universal homeomorphism. It follows that f is separated, quasi-compact, and a universal homeomorphism. See Morphisms of Stacks, Lemma lemma check separated covering (Tag 06TZ), lemma check quasi compact covering (Tag 0DTL), and lemma check universal homeomorphism covering (Tag 0DTQ)

To finish the proof we have to show that $f : \mathcal{X} \to M$ is a uniform categorical moduli space. To prove this it suffices to show that given a flat morphism $M' \to M$ of algebraic spaces, the base change

$$M' \times_M \mathcal{X} \longrightarrow M'$$

is a categorical moduli space. Thus we consider a morphism

$$\theta : M' \times_M \mathcal{X} \longrightarrow E$$

where E is an algebraic space. For each i we know that f_i is a uniform categorical moduli space. Hence we obtain

$$\begin{array}{ccc} M' \times_M \mathcal{X}_i & \longrightarrow & M' \times_M \mathcal{X} \\ \downarrow & & \downarrow \theta \\ M' \times_M M_i & \xrightarrow{\psi_i} & E \end{array}$$

Since $\{M' \times_M M_i \to M'\}$ is an étale covering, to obtain the desired morphism $\psi : M' \to E$ it suffices to show that ψ_i and ψ_j agree over $M' \times_M M_i \times_M M_j = M' \times_M M_{ij}$. This follows easily from the fact

that $f_{ij} : \mathcal{X}_{ij} = \mathcal{X}_i \times_{\mathcal{X}} \mathcal{X}_j \to M_{ij}$ is a uniform categorical quotient; details omitted. Then finally one shows that ψ fits into the commutative diagram

$$\begin{array}{ccc} M' \times_M \mathcal{X} & & \\ \downarrow & \searrow^{\theta} & \\ M' & \xrightarrow{\psi} & E \end{array}$$

because "$\{M' \times_M \mathcal{X}_i \to M' \times_M \mathcal{X}\}$ is an étale covering" and the morphisms ψ_i fit into the corresponding commutative diagrams by construction. This finishes the proof of the Keel-Mori theorem. $\square$